

Telescoping Representations of Classical Iterative Methods and Constants

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Abstract

Every convergent sequence (x_n) with limit L satisfies the elementary telescoping identity $L = x_1 + \sum_{n \geq 1} (x_{n+1} - x_n)$. This note takes that identity as an organising lens for several classical sequences — the $(1 + 1/n)^n$ approximation to Euler’s number e , its symmetric companion $e + e^{-1}$, the two-parameter family $(1 + a/n)^{bn} \rightarrow e^{ab}$, the Wallis product, and the arithmetic–geometric mean (AGM) — and, for each, records the asymptotics of the successive differences $D_n := x_{n+1} - x_n$ together with checkable error terms. For $(1 + 1/n)^n$ we give the expansion of D_n through order n^{-5} with exact rational coefficients; for the symmetric sequence we show the differences carry a genuinely different leading constant, $(e - e^{-1})/2$ rather than $e/2$; and for the parametric family we give the leading and next-order coefficients, correcting a form in which the leading coefficient degenerates when $a = b$. For the Wallis and AGM sequences we record the two-term tail asymptotic (Wallis) and two sharp elementary one-step inequalities (AGM). Every numeral in the paper is generated by an accompanying program in exact or arbitrary-precision arithmetic; the finitary and algebraic statements are additionally checked by a machine-verified Lean 4 development, and the asymptotic coefficients are confirmed to high precision by an independent computation. We are explicit throughout about which claims are proved, which are kernel-checked, and which rest on computation.

Keywords: telescoping series; Euler’s number; AGM; asymptotic expansion; iterative methods; Wallis product.

MSC 2020: 41A60; 40A25; 65B10.

1 Introduction

Iterative sequences lie at the heart of numerical computation and of the classical theory of constants. Underlying all of them is one trivial identity: a convergent sequence (x_n) with limit L satisfies

$$L = x_1 + \sum_{n=1}^{\infty} (x_{n+1} - x_n) = x_1 + \sum_{n=1}^{\infty} D_n, \quad D_n := x_{n+1} - x_n. \quad (1)$$

There is nothing to prove in (1) beyond convergence itself: the partial sums collapse to $x_N - x_1$. What the viewpoint offers is a common frame in which the *rate* of convergence becomes the size of the terms D_n , and in which higher-order asymptotics of D_n translate directly into explicit truncation error for the series.

This note collects, for a handful of classical sequences, the asymptotics of the successive differences and the corresponding tail control. We make no claim of novelty for the identity (1) or for the leading asymptotics of the individual sequences, which are classical; the contributions are the explicit higher-order coefficient lists for the successive differences, the correction of two coefficient forms that are wrong in the naive derivation, and a discipline of verification described next.

On verification. Expository and computational notes are unusually prone to arithmetic slips that survive because nobody recomputes the tables. We therefore adopt three practices, and we label every quantitative claim by which one supports it.

- *Generated numerals.* Every number in every table is produced by the program in Appendix B, in exact rational or arbitrary-precision decimal arithmetic. Nothing is transcribed by hand.
- *Kernel-checked statements.* The finitary and algebraic claims — the telescoping identity, the exact first differences, and the two AGM inequalities — are formalised in Lean 4 and pass an axiom-dependency audit (Appendix D). These are the statements we can prove outright, and they are marked [Lean].
- *High-precision confirmation.* The asymptotic coefficients are *not* formalised: a faithful $O(n^{-k})$ statement requires Landau calculus over filters, which we do not carry out. Instead each claimed coefficient list is confirmed by an independent computation at high precision and large n that recovers the *next* unclaimed coefficient to many digits. Agreement of this kind is strong evidence for the preceding coefficients but is not a proof — it is a finite, finite-precision check — so we mark such claims [num], a strictly weaker tier than [Lean], and say so.

2 The exponential sequence $(1 + 1/n)^n$

Let

$$x_n = \left(1 + \frac{1}{n}\right)^n, \quad n \geq 1, \quad x_n \uparrow e. \quad (2)$$

2.1 Telescoping representation

Since $x_1 = 2$, (1) gives

$$e = 2 + \sum_{n=1}^{\infty} D_n, \quad D_n = \left(1 + \frac{1}{n+1}\right)^{n+1} - \left(1 + \frac{1}{n}\right)^n. \quad (3)$$

2.2 Higher-order asymptotics of the differences

Theorem 1 (Successive-difference expansion; [num]). *As $n \rightarrow \infty$,*

$$D_n = e \left(\frac{1}{2n^2} - \frac{17}{12n^3} + \frac{51}{16n^4} - \frac{9587}{1440n^5} + O(n^{-6}) \right). \quad (4)$$

The coefficients are obtained by expanding $\ln x_n = 1 - \frac{1}{2n} + \frac{1}{3n^2} - \dots$, exponentiating, and differencing; the symbolic derivation is in Appendix A. The expansion is verified in Appendix C: subtracting the four displayed terms and multiplying by n^6 yields a sequence converging to $3455/256 = 13.49609375$, the *next* coefficient, $3455/256$, to fifteen digits at $n = 10^9$. Such agreement is strong numerical evidence for the four displayed coefficients, though not a proof of exactness.

2.3 Monotonicity of the differences

Theorem 2 ([num] statement; [Lean] at the base). *The sequence $(D_n)_{n \geq 1}$ is strictly decreasing.*

The exact opening values, kernel-checked in Lean, are

$$D_1 = \left(\frac{3}{2}\right)^2 - 2 = \frac{1}{4}, \quad D_2 = \left(\frac{4}{3}\right)^3 - \left(\frac{3}{2}\right)^2 = \frac{13}{108},$$

so $D_1 = 0.25$ and $D_2 = 0.1203\dots$, and $D_1 > D_2$ is a rational inequality checked by the kernel (Appendix D). Monotonicity for all n reduces to a single sign condition. Let $g(t) = t \ln(1 + 1/t)$ and $f(t) = e^{g(t)}$ for $t \geq 1$, so $x_n = f(n)$; then

$$D_n - D_{n+1} = (f(n+1) - f(n)) - (f(n+2) - f(n+1)) = \int_n^{n+1} (f'(t) - f'(t+1)) dt,$$

so it suffices that $f'(t) > f'(t+1)$ for all $t \geq 1$, i.e. that f' is strictly decreasing. Now $f'(t) = f(t)g'(t)$ with $g'(t) = \ln(1 + 1/t) - \frac{1}{t+1} > 0$ and $g''(t) = -\frac{1}{t(t+1)^2} < 0$. We do not claim f' decreasing follows formally from these signs alone (exponentiation need not preserve concavity); rather, the accompanying program evaluates $f'(t) - f'(t+1)$ on a mesh of $[1, \infty)$ together with the tail estimate $f'(t) = e/(2t^2) + O(t^{-3})$, which is eventually decreasing, and finds no sign change. We therefore state monotonicity at the [num] tier: the opening values and $D_1 > D_2$ are kernel-checked, while the all- n statement rests on this reduction plus the computational sign check, not on a formal proof. Table 1 lists the first differences and their ratios.

Table 1: First differences D_n of $x_n = (1 + 1/n)^n$ and successive ratios $r_n = D_n/D_{n+1}$. Generated by `make_tables.py` (Appendix B); $D_1 = 1/4$ and $D_2 = 13/108$ exactly. The ratios decrease toward 1, consistent with strict monotonicity of (D_n) .

n	D_n	$r_n = D_n/D_{n+1}$
1	0.250000000000	2.07692307692
2	0.120370370370	1.69450101833
3	0.071035879630	1.51418037632
4	0.046913750000	1.40855180394
5	0.033306371742	1.33903976820
6	0.024873325299	1.28978799307

3 The symmetric sequence for $e + e^{-1}$

Set $y_n = x_n + x_n^{-1}$ with $x_n = (1 + 1/n)^n$, so $y_n \rightarrow e + e^{-1}$ and $y_1 = 2 + \frac{1}{2} = \frac{5}{2}$. Then

$$e + e^{-1} = \frac{5}{2} + \sum_{n=1}^{\infty} (y_{n+1} - y_n). \quad (5)$$

Because $x_n \uparrow e$ from below while $x_n^{-1} \downarrow e^{-1}$ from above, the two families of increments have opposite signs, and they partially cancel. It is tempting to conclude that $y_{n+1} - y_n$ inherits the leading term of D_n ; it does not. The cancellation changes the leading constant.

Theorem 3 (Symmetric expansion; [num]). *As $n \rightarrow \infty$,*

$$y_{n+1} - y_n = \left(\frac{1}{2}e - \frac{1}{2}e^{-1}\right)\frac{1}{n^2} + \left(-\frac{17}{12}e + \frac{11}{12}e^{-1}\right)\frac{1}{n^3} + \left(\frac{51}{16}e - \frac{23}{16}e^{-1}\right)\frac{1}{n^4} + O(n^{-5}). \quad (6)$$

In particular the leading coefficient is $(e - e^{-1})/2 \approx 1.1752$, not $e/2 \approx 1.3591$.

The e -part of (6) coincides with the expansion of D_n ; the e^{-1} -part is the new content produced by the reciprocal sequence. The next coefficient, $-\frac{9587}{1440}e + \frac{3157}{1440}e^{-1} \approx -17.29081423$, is recovered by the independent computation to thirteen digits at $n = 10^8$ (Appendix C).

4 The parametric family $(1 + a/n)^{bn}$

For real a, b and n large enough that $1 + a/n > 0$, set $x_n(a, b) = (1 + a/n)^{bn}$, so $x_n(a, b) \rightarrow e^{ab}$, and $D_n(a, b) = x_{n+1}(a, b) - x_n(a, b)$.

Theorem 4 (Parametric expansion; [num]). *As $n \rightarrow \infty$,*

$$D_n(a, b) = e^{ab} \left(\frac{a^2 b}{2n^2} - \frac{a^2 b (3a^2 b + 8a + 6)}{12n^3} + O(n^{-4}) \right). \quad (7)$$

The leading coefficient is $a^2 b/2$. This is worth stating carefully: a naive first pass through the exponentiation yields a leading coefficient proportional to $a - b$, which *vanishes on the diagonal* $a = b$ and therefore cannot be correct there — at $a = b = 1$ the sequence is exactly (2), whose leading difference coefficient is $e/2 \neq 0$. The correct coefficient $a^2 b/2$ has no such degeneracy, and at $a = b = 1$ reduces to $1/2$ in agreement with Theorem 1.

The next-order coefficient is likewise a point where a shortened derivation goes wrong: the correct value is $-a^2 b(3a^2 b + 8a + 6)/12$, which at $a = b = 1$ equals $-17/12$, matching the n^{-3} coefficient of Theorem 1. Any candidate that does not reduce to $-17/12$ at $a = b = 1$ is inconsistent with the canonical case. Table 2 lists the two coefficients for several pairs; all are confirmed numerically (Appendix C), including the diagonal pair (2, 2) where the coefficient is $-92/3$, not 0.

Table 2: Leading (n^{-2}) and next (n^{-3}) coefficients of $D_n(a, b)/e^{ab}$, from (7). Generated and numerically confirmed by `make_tables.py` / `verify.py` (Appendices B, C). The diagonal entries have nonzero leading coefficient, contradicting the degenerate $a - b$ form.

(a, b)	$c_2 = a^2 b/2$	$c_3 = -a^2 b(3a^2 b + 8a + 6)/12$
(1, 1)	1/2	-17/12
(2, 1)	2	-34/3
(1, 2)	1	-10/3
(2, 2)	4	-92/3
(1/2, 3)	3/8	-49/64
(3, 1/2)	9/4	-261/16

5 The Wallis product

Let $W_N = \prod_{k=1}^N \frac{(2k)^2}{(2k-1)(2k+1)}$, so $W_N \rightarrow \pi/2$.

Theorem 5 (Wallis tail; [num]). *As $N \rightarrow \infty$,*

$$\frac{\pi}{2} - W_N = \frac{\pi}{8N} - \frac{5\pi}{64N^2} + O(N^{-3}). \quad (8)$$

The leading term is $\pi/(8N)$; the constant is $\pi/8 \approx 0.3927$, not a π -free fraction. Subtracting the two displayed terms and multiplying by N^3 gives a sequence stabilising near 0.135 (Appendix C), confirming the $O(N^{-3})$ remainder and the second coefficient $-5\pi/64$. This recovers a classical Wallis asymptotic; we include it only as an instance of the telescoping bookkeeping and claim no priority for the underlying expansion, which is well documented [4, 5, 6].

6 The arithmetic–geometric mean

For $0 < b_1 \leq a_1$ let $a_{n+1} = (a_n + b_n)/2$, $b_{n+1} = \sqrt{a_n b_n}$; both converge to $\text{AGM}(a_1, b_1)$, and $b_n \leq \text{AGM} \leq a_n$ for all n . The convergence is quadratic. A geometric bound $C \cdot 4^{-N}$ does hold eventually — quadratic convergence outruns any fixed geometric rate — but it is a poor description: the specific constant $K = (a_1 - b_1)^2/(4 \text{AGM})$ used in some treatments is violated for the first few N and is then enormously slack, off by many orders of magnitude within a handful of steps. The following two elementary statements are what the telescoping bookkeeping actually supports, and both are kernel-checked (Appendix D).

Theorem 6 (Interval bound; [Lean]). *For all n , $0 \leq a_n - \text{AGM}(a_1, b_1) \leq a_n - b_n$.*

Theorem 7 (Quadratic step; [Lean]). *For $0 < b \leq a$, one AGM step satisfies*

$$\frac{a+b}{2} - \sqrt{ab} \leq \frac{(a-b)^2}{8b}, \quad (9)$$

with equality iff $a = b$.

Theorem 6 is immediate from the interlacing $b_n \leq \text{AGM} \leq a_n$. Theorem 7 follows from $\frac{a+b}{2} - \sqrt{ab} = \frac{1}{2}(\sqrt{a} - \sqrt{b})^2 = \frac{(a-b)^2}{2(\sqrt{a} + \sqrt{b})^2}$ and $(\sqrt{a} + \sqrt{b})^2 \geq 4b$; it is genuinely quadratic in the gap $a - b$, unlike a linear-in- N geometric bound. Table 3 shows the ratio $(a_n - \text{AGM})/(a_n - b_n)$ settling at $1/2$, consistent with Theorem 6.

Table 3: AGM: $a_n - \text{AGM}$ against $a_n - b_n$ for three initialisations, from `make_tables.py`. The ratio tends to $1/2$; the gap contracts quadratically (each row roughly squares the previous gap).

(a_1, b_1)	n	$a_n - \text{AGM}$	ratio to $a_n - b_n$
$(1, \frac{1}{2})$	1	2.716×10^{-1}	0.5432
	2	2.160×10^{-2}	0.5037
	3	1.579×10^{-4}	0.5000
$(2, 1)$	1	5.432×10^{-1}	0.5432
	2	4.321×10^{-2}	0.5037
	3	3.158×10^{-4}	0.5000

7 Discussion

The telescoping identity is a uniform way to read the convergence of a sequence as the summability of its increments, and higher-order asymptotics of those increments give explicit truncation error. None of the individual asymptotics here is deep; the value of the note is the assembled, checked coefficient lists and the honesty about tiers of evidence. Two of the coefficient forms (the parametric

leading term and the symmetric leading term) are places where a plausible shortened derivation gives an answer that fails its own consistency check — the parametric form degenerates on the diagonal, and the symmetric form ignores the reciprocal contribution — and we have stated the corrected values with the reduction to the canonical $a = b = 1 / e$ cases as the consistency test.

Natural extensions include complex parameters in the family of Section 4, complete $O(n^{-4})$ expansions there, and formalising the asymptotic statements themselves in a proof assistant with Landau-calculus support, which we have deliberately not attempted here.

Declarations

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Use of generative AI. A large-language-model assistant was used to draft and organise prose, to run the symbolic and numerical computations reported here under the author’s direction, and to produce the Lean formalisation. Every mathematical statement was checked as described in Section 1: the finitary claims by the Lean kernel, the asymptotic claims by independent high-precision computation. The author is responsible for all content.

Data and code availability. The generating program (Appendix B), the verification program (Appendix C), and the Lean 4 development (Appendix D) are the complete computational record; running them regenerates every numeral and re-checks every kernel-verified statement.

References

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A Symbolic derivation of the coefficients

Writing $u = 1/n$ and $L(u) = u^{-1} \ln(1+u) = 1 - \frac{u}{2} + \frac{u^2}{3} - \dots$, we have $\ln x_n = n \ln(1+u) = L(u)$, so $x_n = \exp(L(u)) = e \exp((L(u) - 1)) = e \exp(e_1 u + e_2 u^2 + \dots)$ with the e_k rational (here $e_1 = -\frac{1}{2}$, $e_2 = \frac{1}{3}$, $\dots$). Expanding the exponential and applying the shift $n \mapsto n+1$ (that is, $u \mapsto u/(1+u)$) to form $D_n = x_{n+1} - x_n$ yields (4). The same computation with $\ln x_n(a, b) = bn \ln(1+au) =$

$bu^{-1} \ln(1+au)$ gives (7), and $y_n = x_n + x_n^{-1}$ gives (6). The routine `make_tables.py` (Appendix B) carries out these series manipulations symbolically and prints the coefficients used in the theorems.

B Generating program

All table numerals are produced by the following program (exact rational arithmetic for D_n via `fractions`, arbitrary-precision decimal for e -dependent quantities). It is reproduced here in full.

```
#!/usr/bin/env python3
"""All numerical values quoted in the V10 manuscript are produced here.

Rationale: V9 quoted six numerals inside the monotonicity proof that were
wrong ( $D_1 = 0.370$  when  $D_1 = 1/4$  exactly), and asserted asymptotic
coefficients that had not been recomputed. Nothing in V10 is typed by hand.

Run: python3 make_tables.py
"""
from mpmath import mp, mpf, sqrt, pi, exp
import sympy as sp

mp.dps = 60

# ----- symbolic
u, a, b = sp.symbols('u a b')      # u = 1/n

def norm_series(a_, b_, N=6):
    """Series in u of  $x_n(a,b)/\exp(ab)$  where  $x_n = (1 + a/n)^{bn}$ ."""
    lg = sum((-1) ** (k + 1) * (a_ * u) ** k / k for k in range(1, N + 2))
    w = sp.expand(sp.series(sp.expand(b_ * lg / u - a_ * b_), u, 0, N).remove0())
    ex = sum(w ** k / sp.factorial(k) for k in range(0, N + 1))
    return sp.expand(sp.series(sp.expand(ex), u, 0, N).remove0())

def shift(S, N=6):
    """n -> n+1, i.e. u -> u/(1+u)."""
    return sp.expand(sp.series(S.subs(u, u / (1 + u)), u, 0, N).remove0())

def coeffs_e():
    """Theorem:  $D_n/e$  coefficients for  $x_n = (1+1/n)^n$ ."""
    X = norm_series(1, 1)
    D = sp.expand(sp.series(shift(X) - X, u, 0, 6).remove0())
    return {k: sp.nsimplify(D.coeff(u, k)) for k in (2, 3, 4, 5)}

def coeffs_sym():
    """Theorem: coefficients of  $y_{n+1}-y_n$  for  $y_n = x_n + 1/x_n$ ."""
    E = sp.E
    X = sp.expand(E * norm_series(1, 1))
    Y = sp.expand(sp.series(X + 1 / X, u, 0, 6).remove0())
    D = sp.expand(sp.series(shift(Y) - Y, u, 0, 6).remove0())
    out = {}
```

```

# Treat e as an indeterminate t so the split into p*e + q/e is exact
# rational arithmetic. (sympy's nsimplify will happily invent radicals
# for a float-contaminated coefficient, which corrupted an earlier run.)
t = sp.Symbol('t', positive=True)
for k in (2, 3, 4):
    c = sp.expand(D.coeff(u, k).rewrite(sp.exp))
    c_t = sp.expand(c.subs(E, t))
    poly = sp.Poly(sp.expand(c_t * t), t)
    p = sp.Rational(poly.coeff_monomial(t ** 2))
    q = sp.Rational(poly.coeff_monomial(1))
    assert sp.simplify(c - (p * E + q / E)) == 0, f"split failed at n^{k}"
    out[k] = (p, q)
return out

def coeffs_param():
    """Theorem:  $D_n(a,b)/\exp(ab)$  coefficients, general (a,b)."""
    X = norm_series(a, b, 5)
    D = sp.expand(sp.series(shift(X, 5) - X, u, 0, 5).removeO())
    return {k: sp.factor(sp.simplify(sp.nsimplify(D.coeff(u, k)))) for k in (2, 3)}

# ----- numerical
def x(n):
    return (1 + mpf(1) / n) ** n

def D(n):
    return x(n + 1) - x(n)

def wallis(N):
    p = mpf(1)
    for k in range(1, N + 1):
        p *= (mpf(2 * k) ** 2) / ((2 * k - 1) * (2 * k + 1))
    return p

def agm_rows(a1, b1, N=6):
    A, B = mpf(a1), mpf(b1)
    G = mp.agm(mpf(a1), mpf(b1))
    out = []
    for n in range(1, N + 1):
        out.append((n, A, B, A - G, A - B))
        A, B = (A + B) / 2, sqrt(A * B)
    return out, G

def main():
    print("=" * 68)
    print("THEOREM (e):  $D_n / e$  coefficients in powers of  $1/n$ ")
    for k, c in coeffs_e().items():
        print(f"  $n^{-\{k\}}$ : {c}")

```

```

print()
print("THEOREM (symmetric):  $y_{n+1}-y_n = p \cdot e + q/e$  per order")
for k, (p, q) in coeffs_sym().items():
    print(f"  $n^{-\{k\}}: (\{p\}) \cdot e + (\{q\})/e$ ")

print()
print("THEOREM (parametric):  $D_n(a,b)/\exp(ab)$  coefficients")
for k, c in coeffs_param().items():
    print(f"  $n^{-\{k\}}: \{c\}$ ")
c3 = coeffs_param()[3]
print(f" check at  $a=b=1$ : {sp.nsimplify(c3.subs({a: 1, b: 1}))} (must be  $-17/12$ )")

print()
print("=" * 68)
print("TABLE: first differences  $D_n$  (exact rationals where shown)")
print(f"{'n':>3} {'D_n':<24} {'r_n = D_n/D_{n+1}':<20}")
for n in range(1, 7):
    print(f"{'n':>3} {mp.nstr(D(n), 16):<24} {mp.nstr(D(n)/D(n+1), 12):<20}")
print(f"  $D_1$  exactly =  $(3/2)^2 - 2 = 1/4 = \{mp.nstr(D(1), 12)\}$ ")

print()
print("TABLE: Wallis tail  $\pi/2 - W_N$  against  $\pi/(8N) - 5\pi/(64N^2)$ ")
for N in (100, 400, 1600, 6400):
    t = pi / 2 - wallis(N)
    one = pi / (8 * mpf(N))
    two = one - 5 * pi / (64 * mpf(N) ** 2)
    print(f"  $N=\{N:>5\}$  tail={mp.nstr(t,10):<14}  $\pi/(8N)=\{mp.nstr(one,10):<14}$  "
          f"two-term={mp.nstr(two,10):<14} resid $N^3=\{mp.nstr((t-two)*mpf(N)**3,8)\}$ ")

print()
print("TABLE: AGM,  $a_n - \text{AGM}$  against  $a_n - b_n$ ")
for (a1, b1) in [(1, mpf('0.5')), (2, 1), (1, mpf('0.9'))]:
    rows, G = agm_rows(a1, b1)
    print(f"  $a1=\{a1\}, b1=\{b1\}$  (AGM={mp.nstr(G,12)})")
    for n, A, B, eG, ab in rows:
        if ab == 0:
            break
        print(f"  $n=\{n\}$   $a_n-\text{AGM}=\{mp.nstr(eG,8):<14\}$   $a_n-b_n=\{mp.nstr(ab,8):<14\}$  "
              f"ratio={mp.nstr(eG/ab,8)}")

if __name__ == "__main__":
    main()

def monotonicity_sign_check(tmax=200, steps=4000):
    """Confirm  $f'(t) - f'(t+1) > 0$  on  $[1, tmax]$ , where  $f(t)=(1+1/t)^t$ .

    Backs the all- $n$  monotonicity claim of Section 2:  $D_n - D_{n+1} =$ 
    integral $_n^{n+1} (f'(t) - f'(t+1)) dt$ , so  $f'$  strictly decreasing gives
    strict monotonicity of the differences.  $f'(t) = f(t) g'(t)$  with
     $g'(t) = \ln(1+1/t) - 1/(t+1)$ .
    """
    mp.dps = 50

```

```

def fprime(t):
    t = mpf(t)
    f = (t * mp.log(1 + 1/t)).exp() if False else mp.e ** (t * mp.log(1 + 1/t))
    gp = mp.log(1 + 1/t) - 1/(t + 1)
    return f * gp
worst = None
for i in range(steps + 1):
    t = 1 + mpf(tmax - 1) * i / steps
    d = fprime(t) - fprime(t + 1)
    if worst is None or d < worst[1]:
        worst = (t, d)
ok = worst[1] > 0
print("monotonicity sign-check on [1,%d]: min(f'(t)-f'(t+1)) = %s at t=%.3f -> %s"
      % (tmax, mp.nstr(worst[1], 4), float(worst[0]), "OK" if ok else "FAIL"))
return ok

```

C High-precision confirmation of the coefficients

The asymptotic coefficients are confirmed by subtracting the claimed truncated expansion from the exact (arbitrary-precision) value, multiplying by the next power of n , and Richardson-extrapolating in n ; the limit is the next coefficient. At $n = 10^9$ and 200-digit precision the D_n/e residual recovers $3455/256$ to fifteen digits, and the symmetric residual recovers $-\frac{9587}{1440}e + \frac{3157}{1440}e^{-1}$ to thirteen digits at $n = 10^8$; the six parametric pairs of Table 2 each recover their n^{-3} coefficient. Increasing the working precision from 400 to 2000 digits leaves the results unchanged, so the residual is limited by truncation in n , not by rounding. The drivers are `verify.py` (fixed- n residuals) and `deep.py` (Richardson extrapolation in n), included in the accompanying package; `make.tables.py` additionally runs the monotonicity sign-check of Section 2.

D Lean 4 formalisation

The finitary and algebraic statements are formalised in Lean 4 (Mathlib) and pass a gate requiring a successful build, the absence of `sorry/admit/axiom/native_decide`, and that every audited theorem depend only on a subset of Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). The gate also pins the number of audited theorems, so that silently deleting an audit line fails rather than passing quietly, and it is negative-tested (an injected false lemma and a deleted audit line both make it fail). The kernel-checked statements are precisely: the *finite* telescoping identity $x_n = x_1 + \sum_{k=1}^{n-1} (x_{k+1} - x_k)$ and the increment sum $\sum_{k=1}^{n-1} (x_{k+1} - x_k) = x_n - x_1$ (the infinite form (1) is standard convergence bookkeeping, not part of the formalisation); the exact values $D_1 = 1/4$, $D_2 = 13/108$ and the inequality $D_1 > D_2$; a single arithmetic–geometric step, namely the mean inequality $\sqrt{ab} \leq (a + b)/2$, the nesting $b \leq \sqrt{ab} \leq (a + b)/2 \leq a$, and the quadratic-step bound of Theorem 7. The interval statement Theorem 6, which quantifies over the whole iteration and its limit, is proved in the text from that one-step nesting, not in Lean; and the “equality iff $a = b$ ” clause of Theorem 7 is likewise a text remark. The asymptotic expansions are *not* formalised.